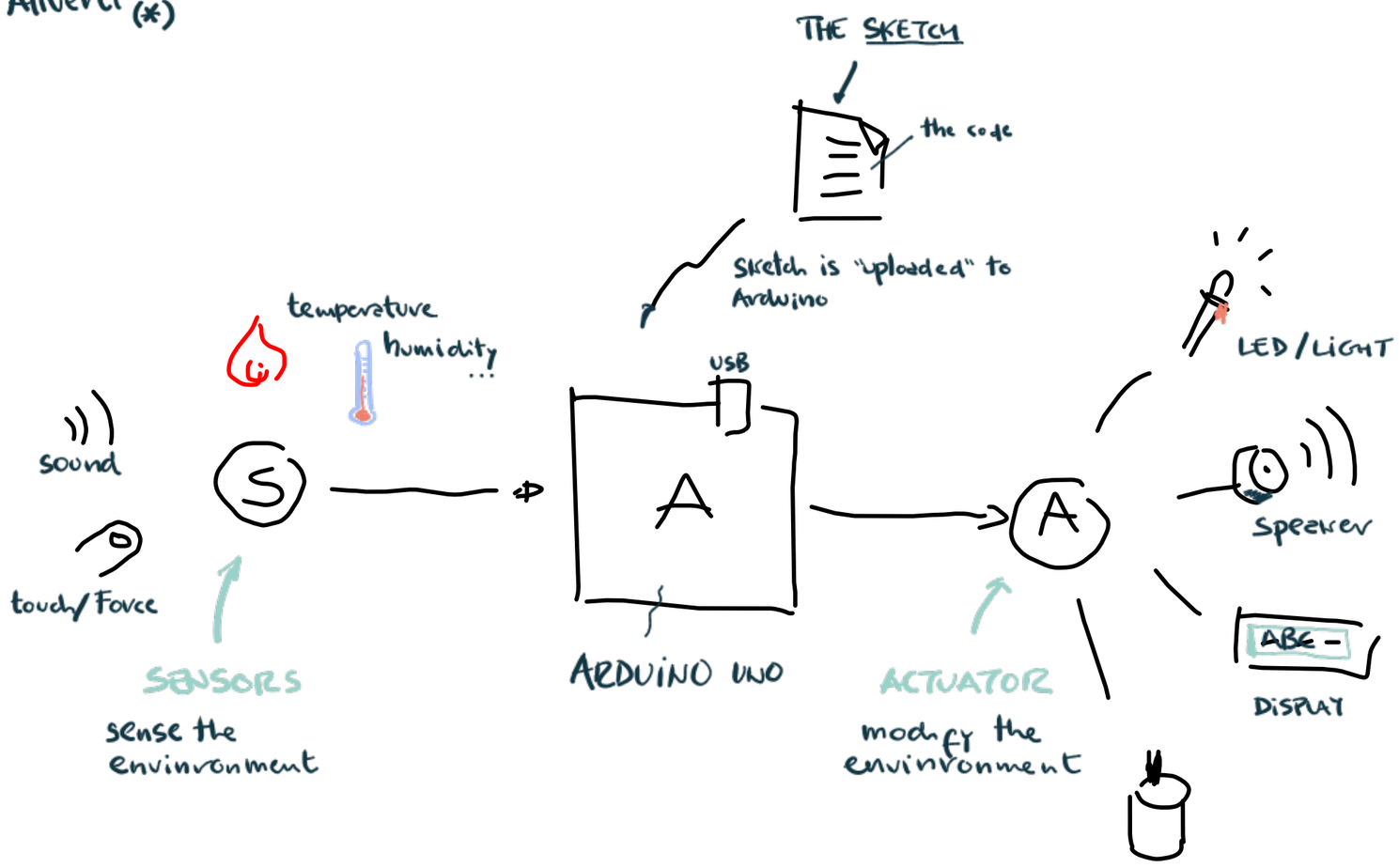
02.03.2026

ARDUINO @ IED

by Paolo Aliverti (*)



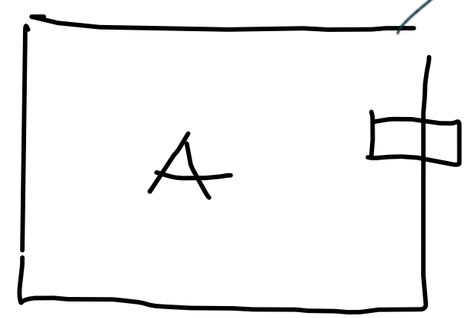
(*) MORE INFO AND FREE TUTORIALS ON MY CHANNEL - LOOK FOR "ALIVERTI PAOLO" — AND THE ARDUINO PLAY LIST -

ARDUINO IDE

is the tool to write the code for Arduino

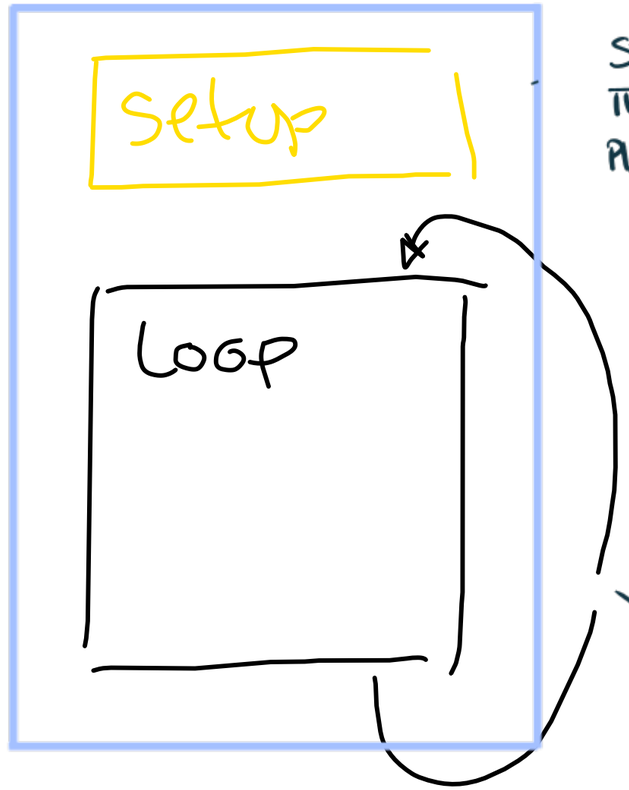
download from → www.arduino.cc

sketch



once "downloaded" into the board, it remains also if you turn off the board

STRUCTURE OF A SKETCH



SETUP

The code here it's executed at power on

Place here configurations, pin settings and so on...

LOOP

The code here is repeated forever... until the board is powered

the loop is fast

16'000'000 operations per second.

EX. 1 – TURN ON PIN

SETUP:

use: `pinMode(10, OUTPUT);`

pin #

LOOP

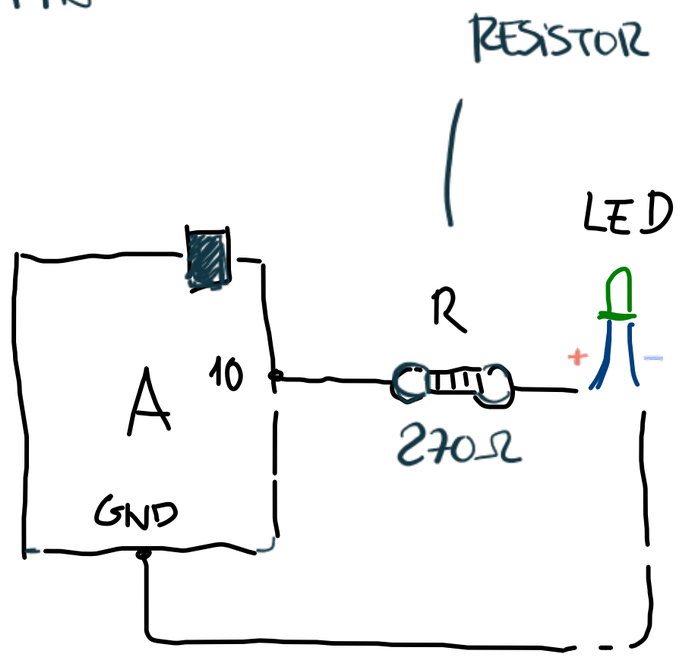
to drive the pin:

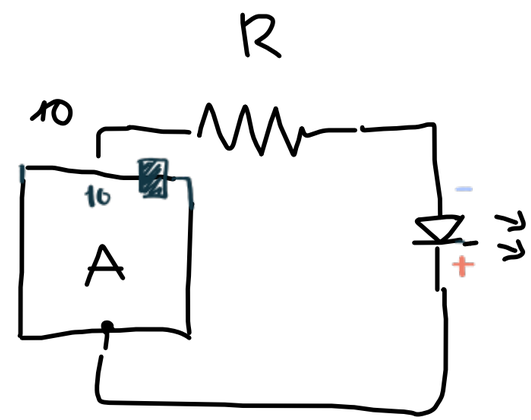
`digitalWrite(10, HIGH);`

PIN ON..

LOW

PIN OFF

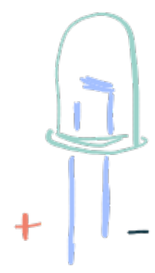




Ω = OHM
unit of measure
of resistance

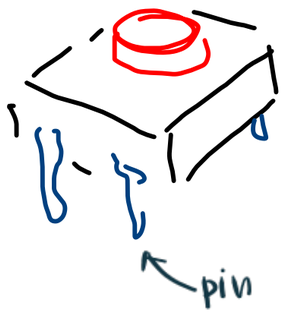
SYMBOL OF A RESISTOR

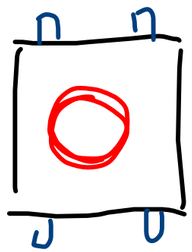
SYMBOL OF A LED

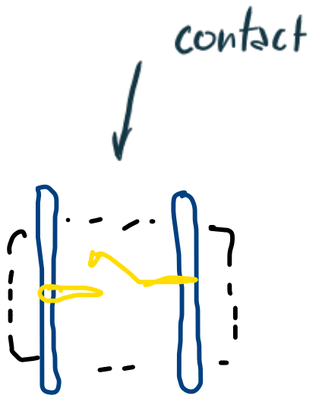


LED

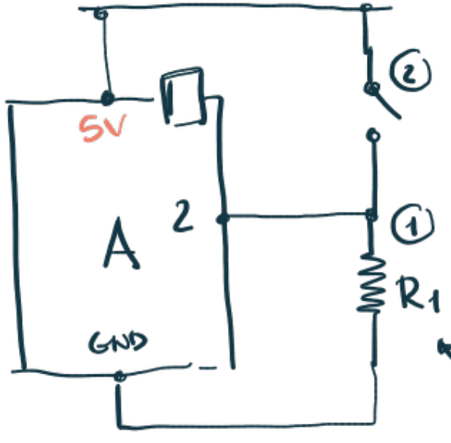
PUSH BUTTON







SYMBOL



① when button isn't pressed the pin #2 is at 0V (GND)

② pressing the button, we connect the pin #2 to 5V

PULL DOWN RESISTOR
tie down the pin #2

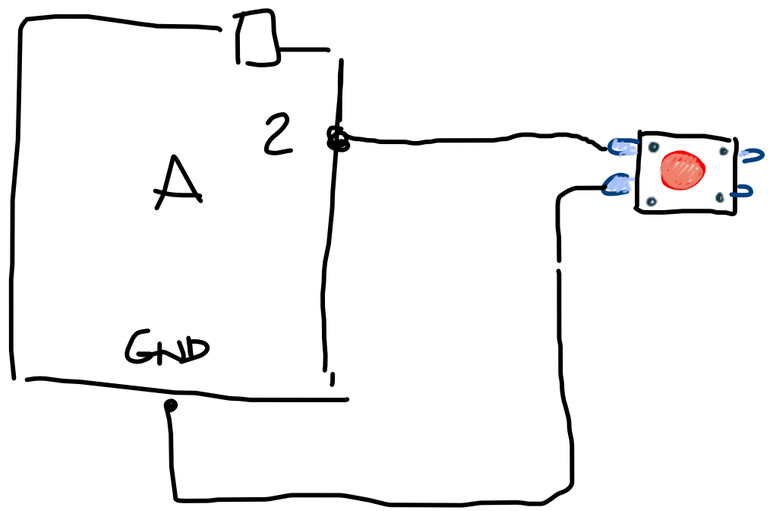
$R_1 = 10'000\,\Omega$ or $10\,k\Omega$

pin #

SETUP : use: pinMode (2, INPUT);

LOOP : use: digitalRead (2);
to read pin state
- HIGH — Button pressed
- LOW — Button not pressed

EX2. - CONNECT A BUTTON



without "PULL DOWN" RESISTOR

- simple wiring
- inside "setup" use: pinMode (2 , INPUT_PULLUP);
- loop : int bt = ! digitalRead (2);

activate internal resistor for the button

if bt == HIGH ⇒ Button pressed

! = NOT

this is a function

function for read pin state

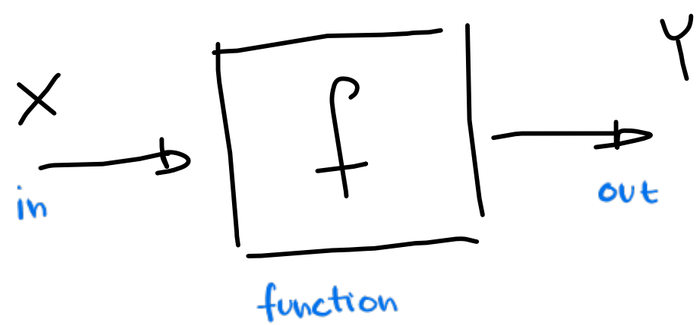
digitalRead (2) → HIGH / LOW

in

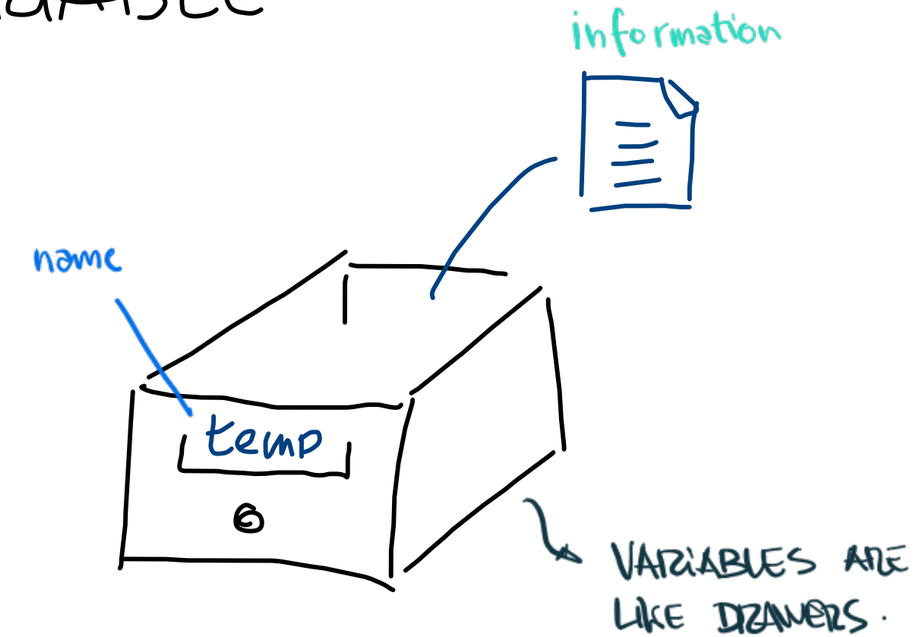
pin number

out...

Returned value



VARIABLE



assign to...

value

```
int temperature = 0;
```

variable type

name

other types:

int	1, 2, 3, -10
float	1,23
double	3,1415.....
long	1'2356789 10'300

IF (expr) {

} else {

}

is sunny?

yes — TRUE

no — FALSE

could be TRUE or FALSE

$t > 12$

button == HIGH

```
if (expression) {
```

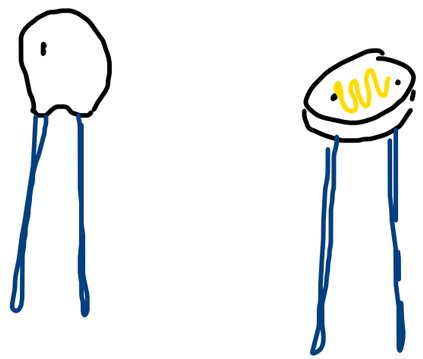
code to execute if expression is true

```
} else {
```

if expression is FALSE... execute this code

```
}
```

SENSORS

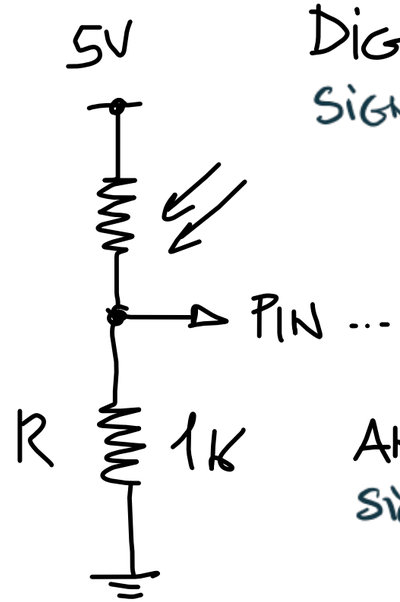


TERMISTORE

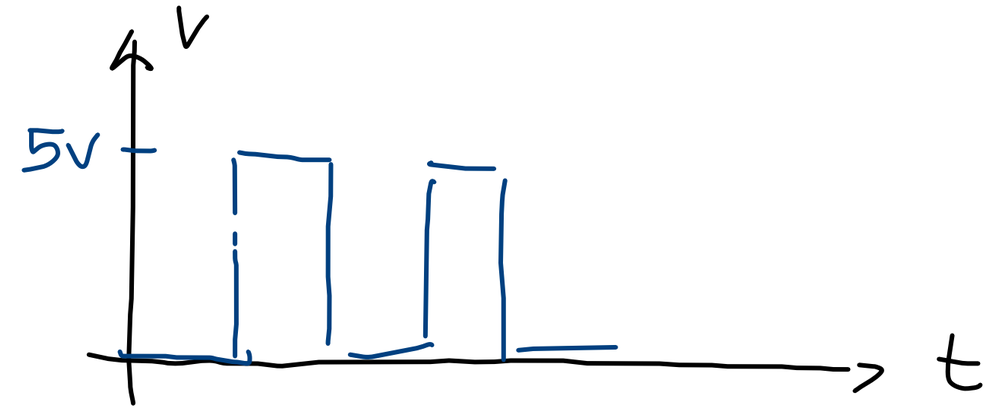
measure temperature

FOTORES.

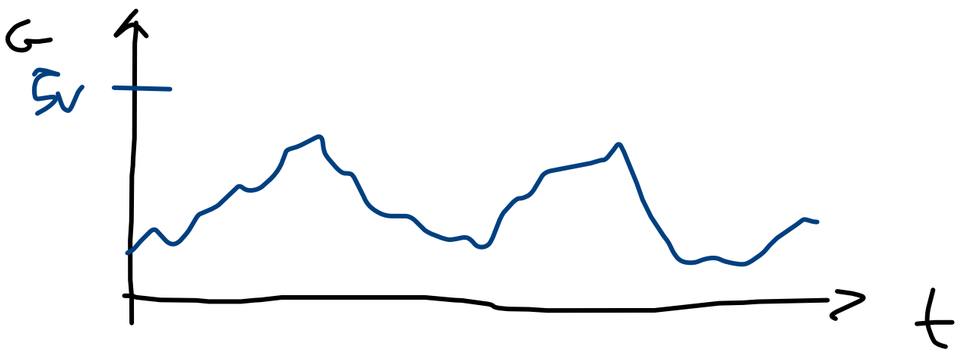
measure light

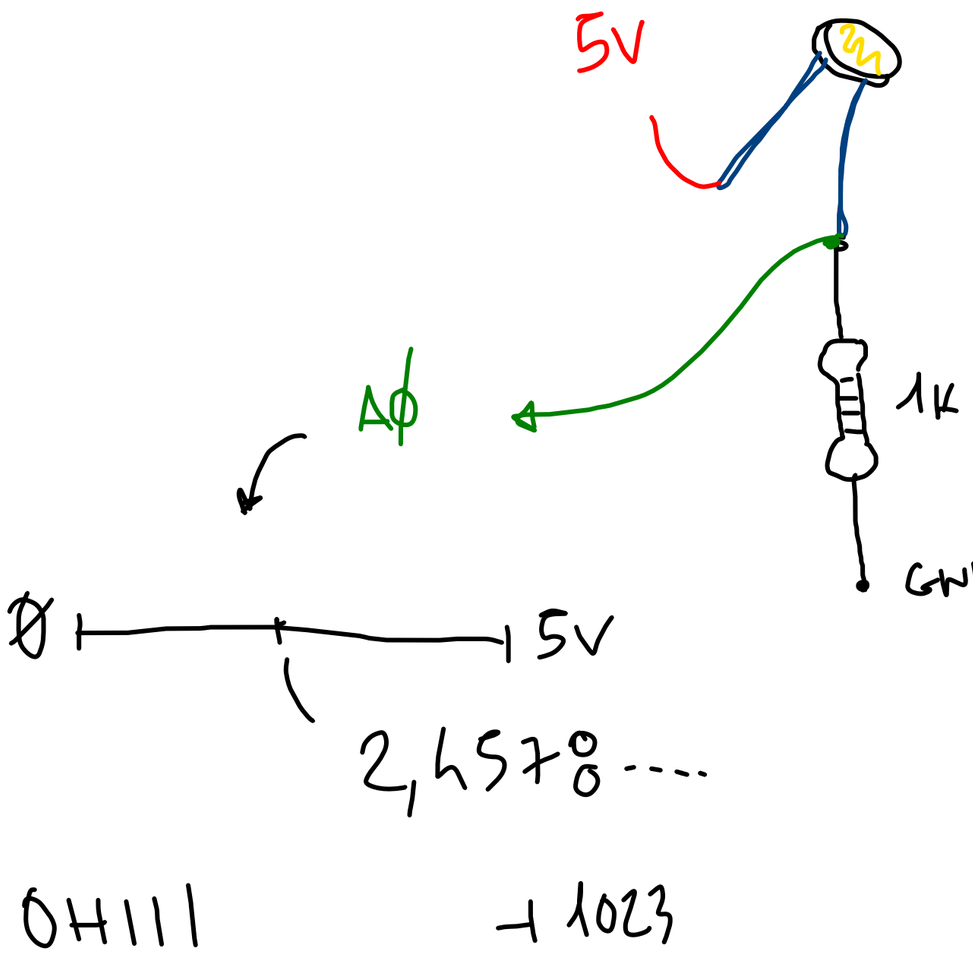


DIGITAL SIGNAL



ANALOG SIGNAL



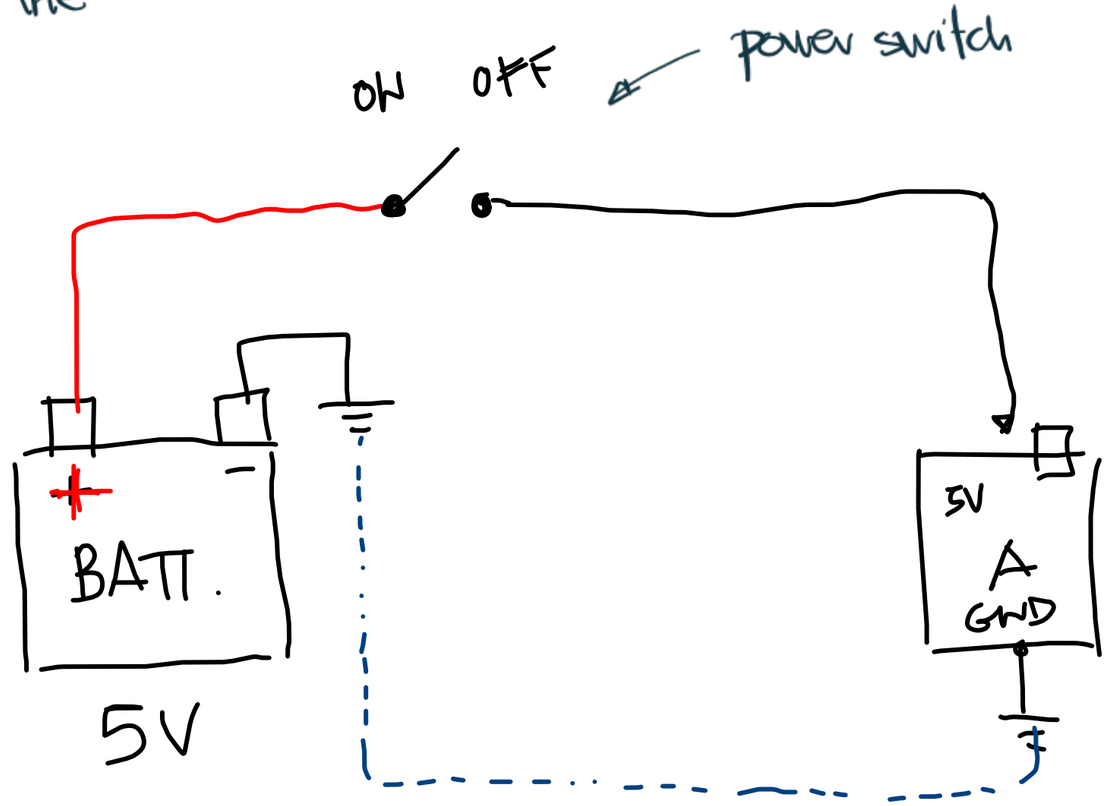


analogRead (A0);

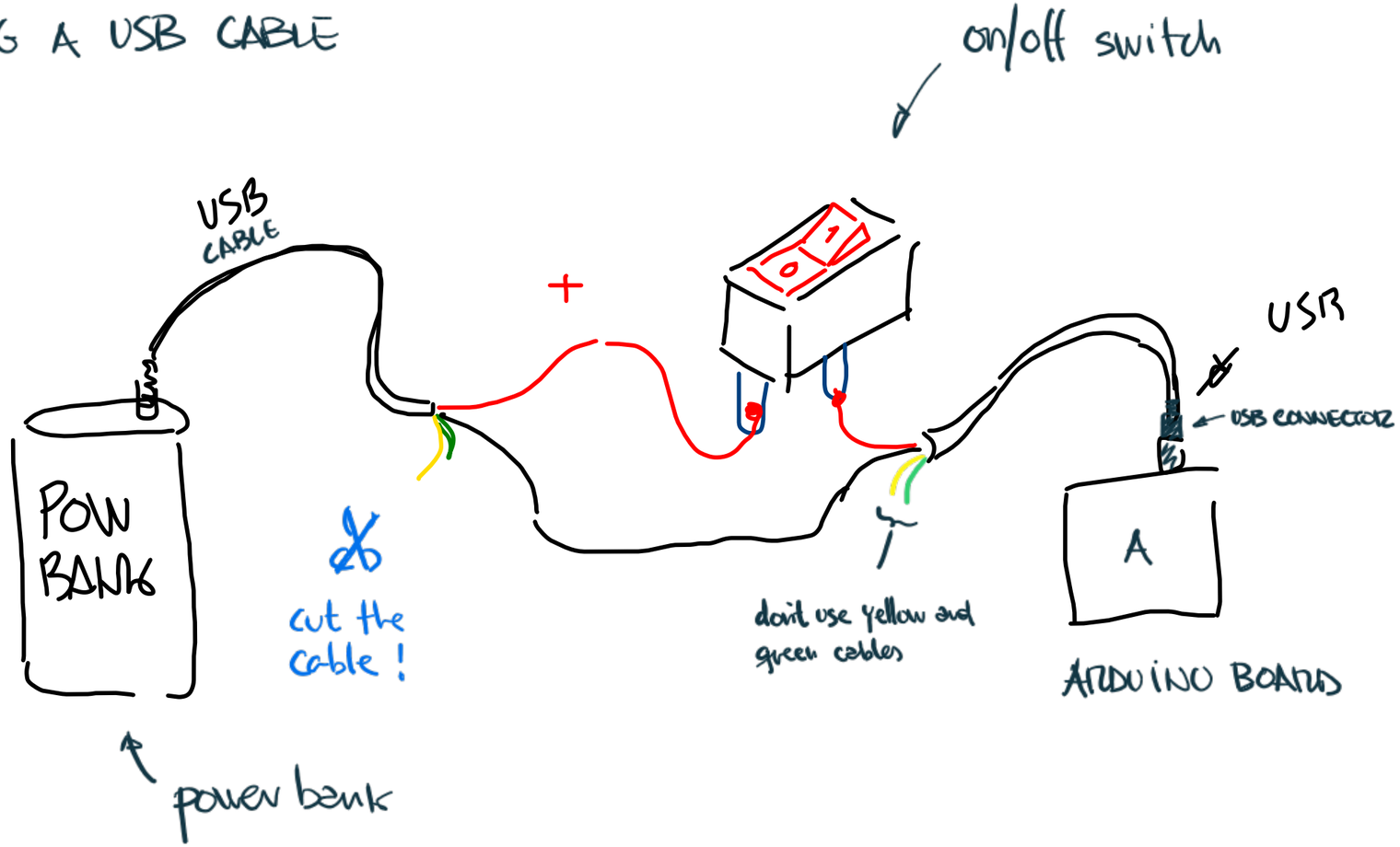
reads the pin A0 and return a number 0 ÷ 1023

0
1
2
⋮
1023

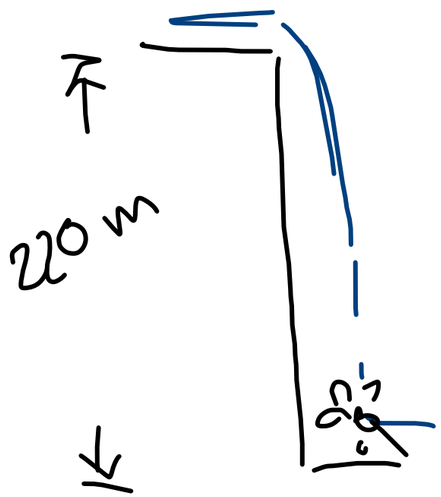
POWER ON THE BOARD



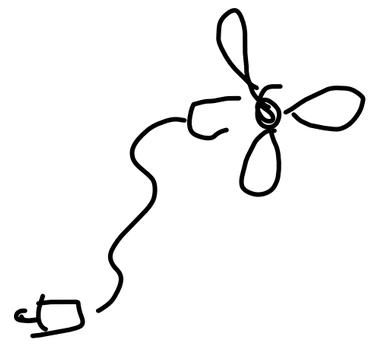
POWER ON USING A USB CABLE



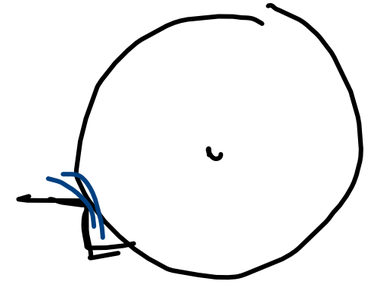
VOLTAGE



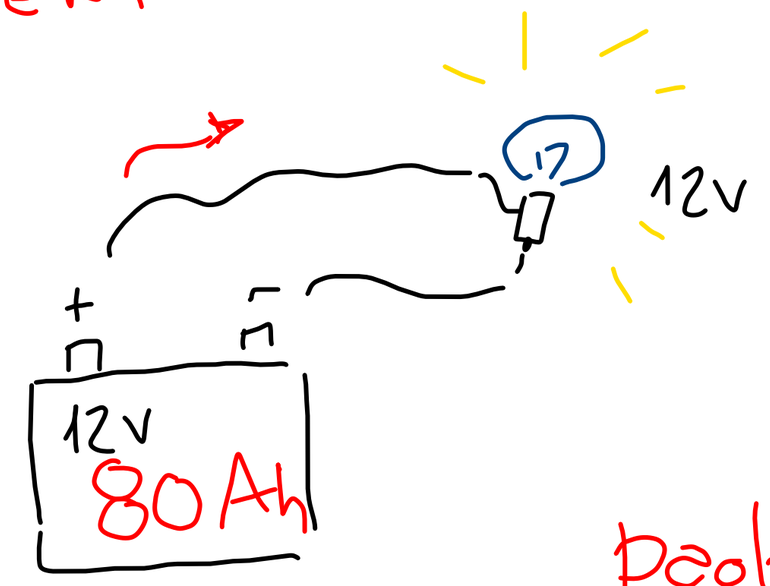
12V



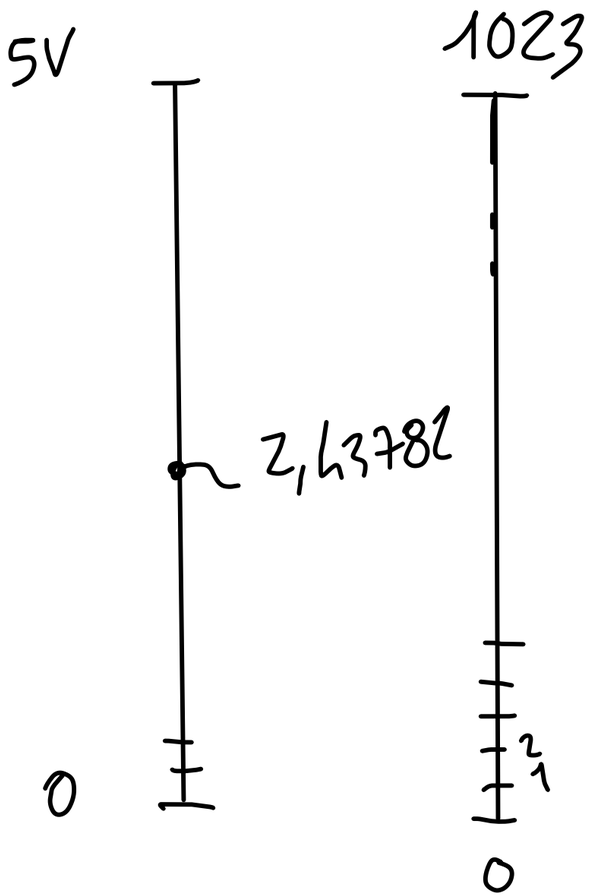
match between water fall and wheel

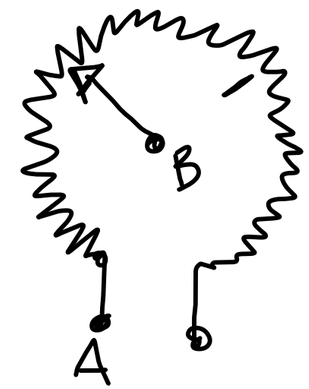


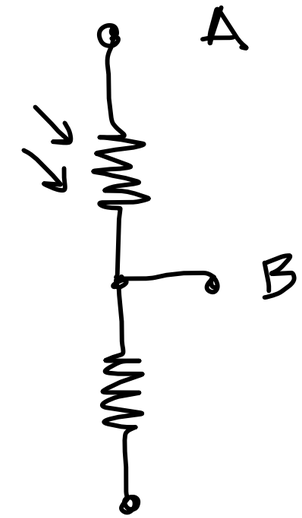
CURRENT
A



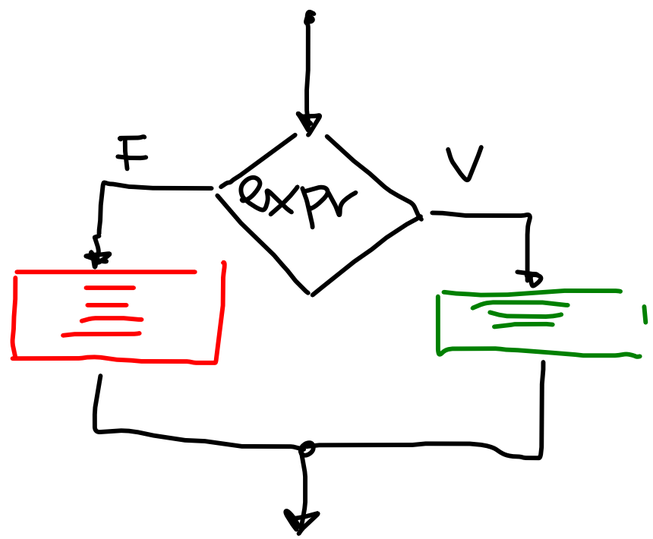
paolo ALIVERTI


5V
1023
2,43781
0
2
1
0


B
A


A
B

SE → IF




```
if (.....) {
    ≡
} else {
    ═
}
```